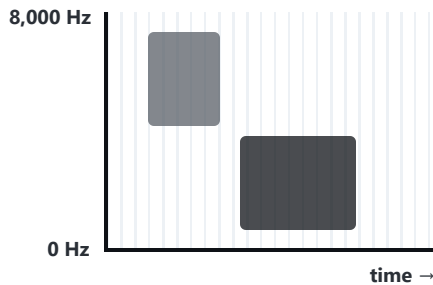


Spectrogram Detective

Before a computer can understand a word, it has to read a picture of the sound. Can you?

How to read a sound picture



- 1 **Time** runs left to right, like reading a sentence.
- 2 **Low sounds** sit at the bottom and **high sounds** at the top (0 to 8,000 Hz, or vibrations per second).
- 3 **Darker** means louder. White means silence.

Detective clues

S Fuzzy static in the top half (above about 4,000 Hz).	SH Fuzzy static that reaches lower than S, darkest near 3,000 Hz.	F Faint fuzz from low to high. Easy to miss!	STOP A blank gap, then a thin line: p, t, k, b, d, g.
EE Stripes with a dark band high up, near 2,000 to 3,000 Hz.	OO Stripes with the dark part squeezed near the bottom.	AH Dark bands close together in the lower middle. The a in "cat" looks similar.	HUM Faint stripes near the bottom: m and n.

Each thin vertical stripe is one buzz of your vocal folds, so vowels and hums look striped. S, SH and F have no buzz, only rushing air, so they look like static. Clue pictures are idealized; real words blend sounds together, so look for the strongest patterns.

1. **Study the clues** with a partner (2 minutes).
2. **Match each card** on the case file to a word from the word bank (8 minutes).
3. **Check the answer key** and explain which clue cracked each case (5 minutes).

Which word is it?

Detective: _____

Partner: _____

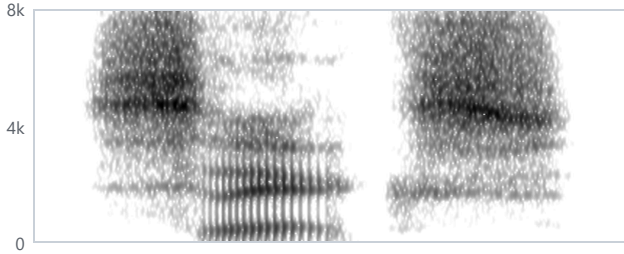
Each picture shows one spoken word. Use the clue sheet to crack all eight cases.

bee · cat · fish · moo · see · shoe · six · sun

Word bank: each word is used exactly once

Card A

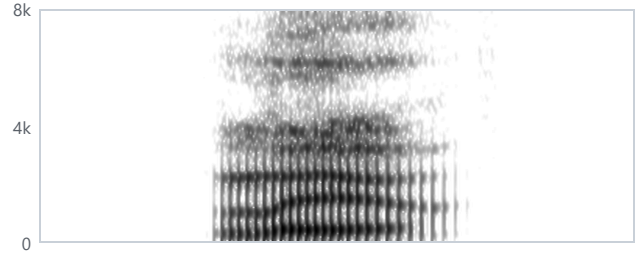
0 to 8,000 Hz · time →



This word is: _____

Card B

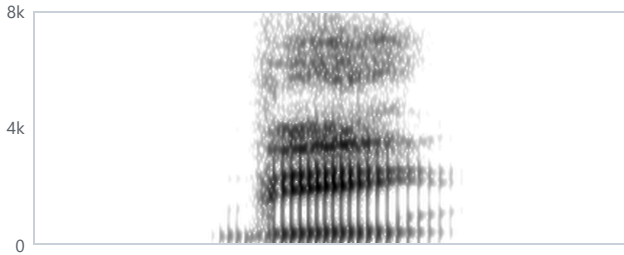
0 to 8,000 Hz · time →



This word is: _____

Card C

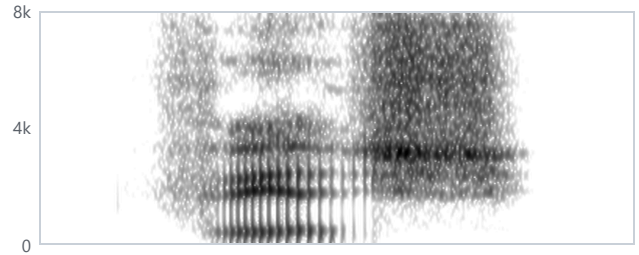
0 to 8,000 Hz · time →



This word is: _____

Card D

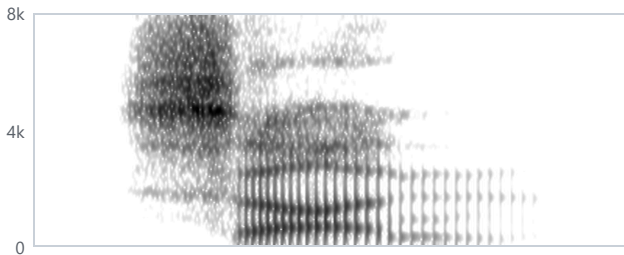
0 to 8,000 Hz · time →



This word is: _____

Card E

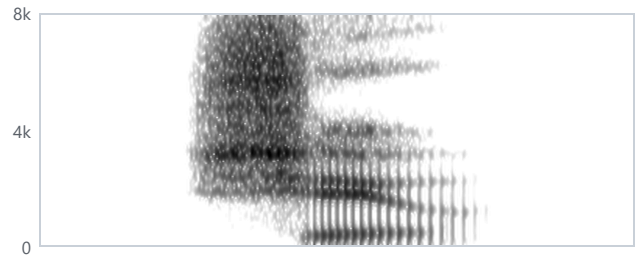
0 to 8,000 Hz · time →



This word is: _____

Card F

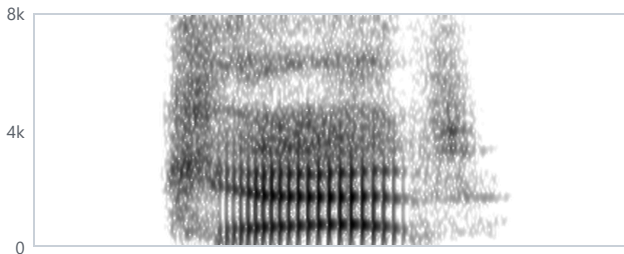
0 to 8,000 Hz · time →



This word is: _____

Card G

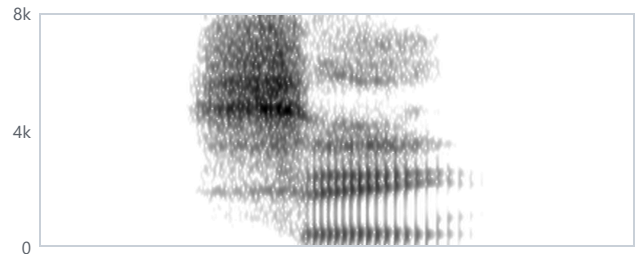
0 to 8,000 Hz · time →



This word is: _____

Card H

0 to 8,000 Hz · time →



This word is: _____

Stuck? Sort first: which cards **start** with static? Which **end** with static? Then look at the vowel bands.

Spectrogram Detective

Students learn to read the same kind of picture a speech recognizer starts from.

TIME

15 minutes

GRADES

3–12

GROUPS

Pairs

MATERIALS

Pages 1–2 (black and white is fine), pencils

Answer key

CARD	WORD	THE CLUE THAT CRACKS IT
A	six	Static at both ends (S ... S), with a short blank gap (the k) before the last one.
B	moo	No static at the start; stripes with the dark part near the bottom (OO).
C	bee	No static at the start; stripes with a dark band up near 2,000 to 3,000 Hz (EE).
D	fish	Faint fuzz first (F), a vowel, then strong static that reaches down toward 2,000 Hz (SH).
E	sun	Static at the top (S), a vowel, then long faint stripes near the bottom (N).
F	shoe	Static that reaches lower than S (SH), then stripes darkest near the bottom (OO).
G	cat	A thin full-height strip first (the k), a vowel, a blank gap, then a thin strip at the end (T).
H	see	Static at the top (S), then stripes with a dark band up near 2,000 to 3,000 Hz (EE).

Fastest solving path: cards A, E, F and H start with static; B, C, D and G do not. Endings separate A from E and D from G, and the vowel bands settle the rest.

Discussion prompts

- Would your voice make the same picture? Children's voices are higher, so the buzz stripes pack closer together and the dark bands sit higher. That is one reason voice assistants find kids harder to understand.
- Why don't computers just follow clues like these? Real speech changes with the speaker, speed and noise. Early systems tried hand-written rules; modern recognizers learn the patterns from many thousands of hours of examples.

Extensions

- **Hard mode:** hide the word bank.
- **Make your own cards:** in the free See Your Voice page from this kit, say a word, press Pause, then Save picture. Swap with a partner and solve each other's cards.
- **AI4K12 connection:** Big Idea 1, Perception (ai4k12.org).

How the pictures were made: word cards use Windows' built-in computer voice, so they are cleaner and slower than real speech. Clue tiles are idealized sounds generated from simple models of the voice. Settings: 8 ms analysis window, 0 to 8,000 Hz, darker = louder.